1. Let r be the distance

$$\sqrt{x^2 + y^2 + (z-2)^2} = r$$

$$\boxed{x^2 + y^2 + (z-2)^2 = r^2}$$

2. Let (x, y, z) be point.

Distance to $z = -1$: $z - (-1) = z + 1$

Distance to $(0,0,1)$: $\sqrt{x^2 + y^2 + (z-1)^2}$

$$\sqrt{x^2 + y^2 + (z-1)^2} = z + 1$$

$$x^2 + y^2 + z^2 - 2z + 1 = z^2 + 2z + 1$$

$$6\left(z + \frac{1}{2}\right) = x^2 + y^2$$

Paraboloid

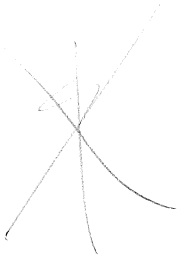
$$z = \frac{x^2 + y^2}{6} - \frac{1}{2}$$

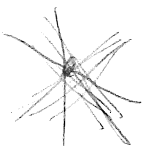
$$\boxed{x^2 + y^2 = 4z}$$

3a) trace $x=0$: $y^2 - z^2 = 0 \Rightarrow y = \pm z$

trace $y=0$: $x^2 - z^2 = 0 \Rightarrow x = \pm z$

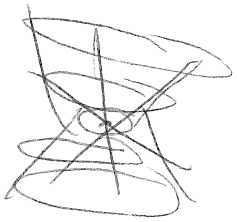
trace $z=0$: $x^2 + y^2 = 0 \Rightarrow$ ~~$x^2 - y^2$~~ $x=0, y=0$





$x=3$: $9 + y^2 - z^2 = 0$

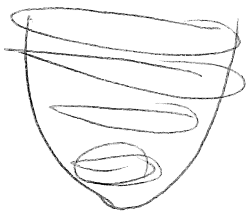
$z^2 - y^2 = 9$

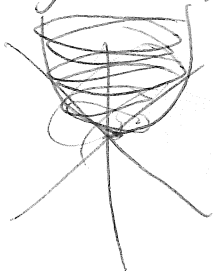


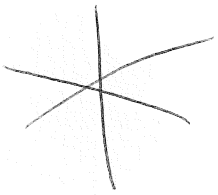
Cones

b) $x=0$: $y^2 = z$

$y=0$: $x^2 = z$ Cute







c) $x^2 - y^2 - z = 0$

$x=0$: $-y^2 - z = 0$

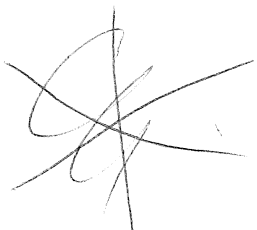
~~$y^2 = z$~~

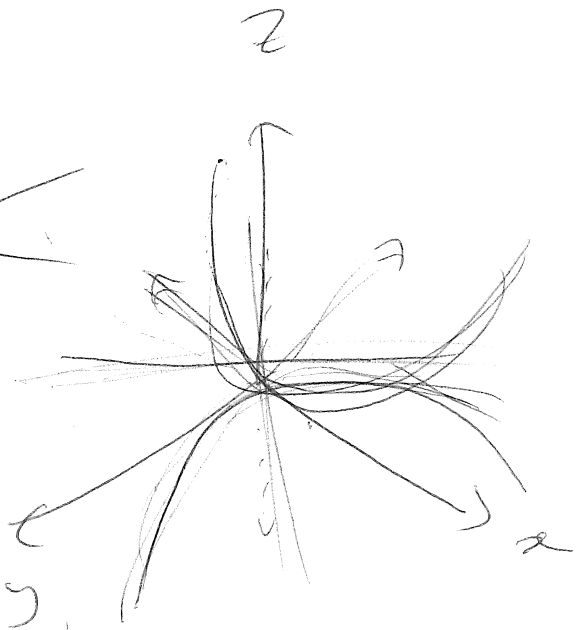
$y=0$: $x^2 - z = 0$

$z = x^2$

$z=0$: $x^2 - y^2 = 0$

$x = \pm y$







$x^2 - y^2 = z$

$z > 0$: $x^2 - y^2 = (\sqrt{z})^2$

$z < 0$: $x^2 - y^2 = -|z|$

$|z| = y^2 - x^2$

$(\sqrt{|z|})^2 = y^2 - x^2$

d) $x^2+y^2+z^2=4$

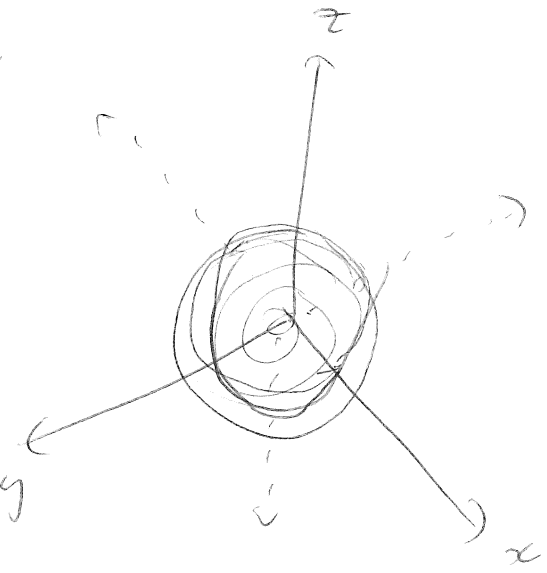
$x=0: \; y^2+z^2=4$

$y=0: \; x^2+z^2=4$

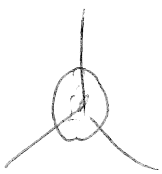
$z=0: \; x^2+y^2=4$

$x^2+y^2=4-z^2$

$\sqrt{4-z^2}$ radius

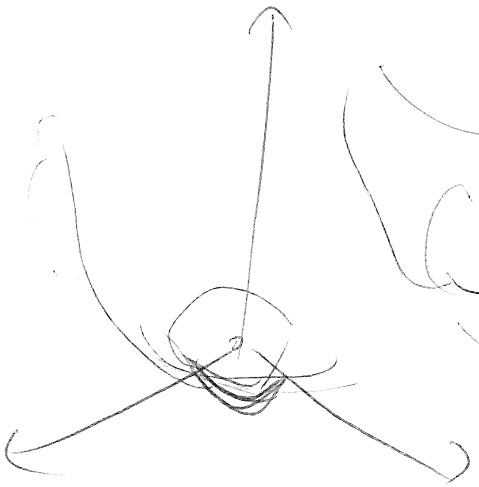


Sphere



e) $x^2+y^2-z^2=4$

$x^2+y^2=4+z^2$

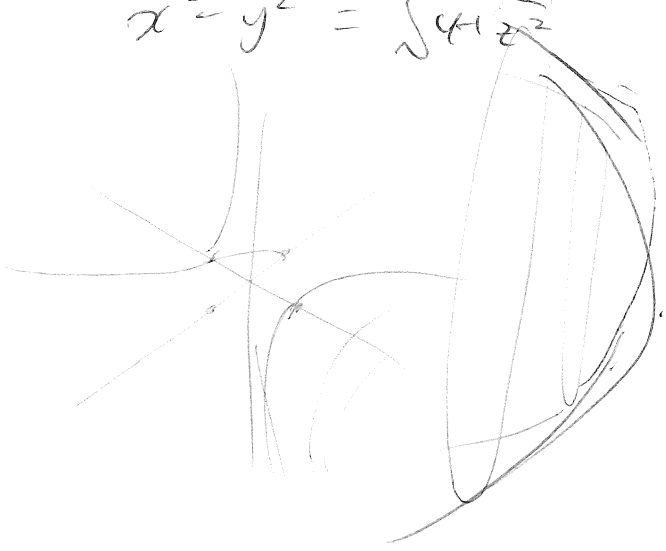




f) $x^2-y^2-z^2=4$

$x^2-y^2=4+z^2$

$x^2-y^2=\sqrt{4+z^2}$



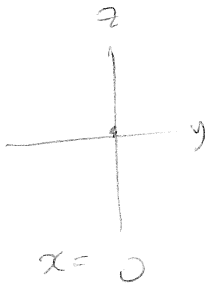


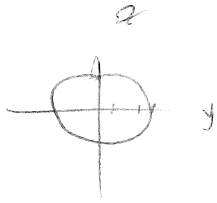
4. $y^2 + z^2 = (4x - x^2)^2$

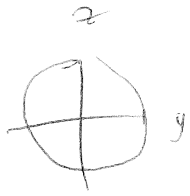
$x = a$: $y^2 + z^2 = (4a - a^2)^2$

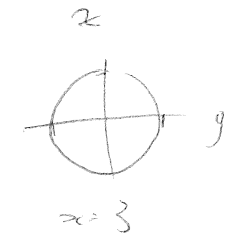
$= a^2(4-a)^2$

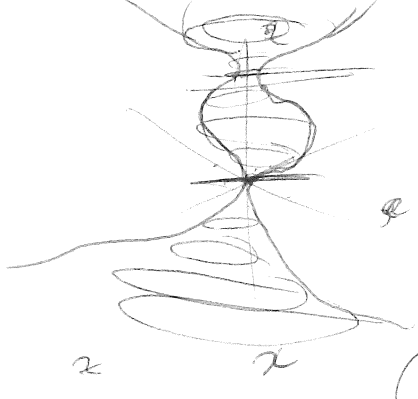
circle







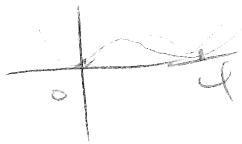




$(4x - x^2)^2$

$(x(4-x))^2$

$x^2(4-x)^2$



Problem 4

(b) Solid of revolution created by revolving $y = 4x - x^2$ about the x-axis. We evaluate the volume between $x = 0$ and $x = 4$.

$$V = \int_0^4 \pi r^2 \, \mathrm{d}x = \pi \int_0^4 \left(4x - x^2\right)^2 \mathrm{d}x = \pi \int_0^4 \left(16x^2 - 8x^3 + x^4\right) \mathrm{d}x = \pi \left[\frac{16x^3}{3} - \frac{8x^4}{4} + \frac{x^5}{5}\right]_0^4$$
$$= \pi \left(\frac{16(4)^3}{3} - 2(4)^4 + \frac{4^5}{5}\right) = \pi \left(\frac{1024}{3} - 512 + \frac{1024}{5}\right) = \pi \left(\frac{5120 - 7680 + 3072}{15}\right) = \boxed{\frac{512\pi}{15}}$$

Problem 5

(a) Solid of revolution formed by rotating $y = \sin z$ about the z axis. Oscillation is in the vertical (z) direction. The traces of the solid parallel to the xy-plane are circles with radius ($\sin z$).

(b) Same as before, in that the traces parallel to the xy-plane are circles, but they are of radius $\sqrt{\sin z}$, which is only real if $\sin z \geq 0 \iff 2\pi k \leq z \leq \pi + 2\pi k$ where $k \in \mathbb{Z}$.

For values of z satisfying $2\pi k \leq z \leq \pi + 2\pi k$ where $k \in \mathbb{Z}$, the trace of the solid parallel to the xy-plane is a circle of radius $\sqrt{\sin z}$. For z-values not satisfying the inequality, the trace at that z value is empty.

The solid is a collection of discontinuous "bulbs" that occur every 2π units in the z-direction.